

CMS-DACO: A Dynamic Ant Colony Optimization Framework for Crowd Evacuation under Fire and Smoke

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Fire Evacuation System*

https://github.com/Sathvikar01/Fire_Evacuation_System

Abstract—Effective building evacuation during fire emergencies demands real-time, adaptive routing that accounts for spreading hazards, smoke, wind, and crowd congestion. This paper presents CMS-DACO, a crowd management system that integrates cellular-automata crowd dynamics with Dynamic Ant Colony Optimization (DACO) for hazard-aware egress. The framework combines a BFS-based distance backbone, annealed ant-based pheromone seeding, dual-channel pheromone for speed and safety, and a predictive congestion model. Fire and smoke are modeled as vectorized fields with wind-biased spread, diffusion, and decay. An asynchronous architecture decouples the simulation backend from the PyQt5 frontend, enabling interactive visualization at scale. Headless Monte-Carlo evaluation on a 20×20 grid (40 agents, 3 exits, wall density 0.08, 200 ticks, $N=30$) demonstrates that DACO achieves 1.00 ± 0.00 evacuation completion compared to 0.10 ± 0.42 for random walk and 1.00 ± 0.00 for distance-greedy baselines, while maintaining low casualty rates and efficient path lengths. The system is fully reproducible with pinned dependencies and deterministic seeding.

Index Terms—crowd simulation, evacuation planning, ant colony optimization, fire dynamics, smoke diffusion, multi-agent systems

I. INTRODUCTION

Rapid and safe evacuation is critical in fire emergencies inside complex buildings. Traditional evacuation models, including cellular automata and social-force systems, capture local movement and bottleneck formation but lack a global learning mechanism that can adapt routes as hazards evolve.

Ant Colony Optimization (ACO), inspired by foraging behavior of ants, provides stigmergic coordination through pheromone trails. When coupled with crowd simulation, ACO enables agents to share implicit knowledge about safe and efficient egress paths. However, integrating ACO with realistic hazard physics remains challenging due to the interplay of discrete crowd motion, continuous fire and smoke fields, and real-time visualization constraints.

We propose CMS-DACO, an open-source framework that addresses these challenges. Our system models evacuation as a discrete grid where agents, fire, smoke, and pheromones co-evolve. The key design principles are (i) realistic hazard physics with wind and diffusion, (ii) balanced heuristic fusion between pheromone, distance, and safety, (iii) congestion-aware routing that anticipates bottlenecks, and (iv) a decoupled

asynchronous execution model that separates compute-intensive simulation from interactive rendering. The framework is built in Python with `numpy` for vectorized physics and `PyQt5` for visualization, and supports both interactive and headless batch evaluation.

The contributions of this work are:

- A unified evacuation model that couples BFS distance fields, annealed pheromone seeding, dual speed/safety channels, and predictive congestion signaling.
- A vectorized fire and smoke model with wind-biased spread, fuel-limited growth, and physically consistent diffusion.
- An asynchronous simulation architecture that achieves interactive rates on grids up to 120×120 while preserving simulation fidelity.
- A reproducible evaluation harness reporting 95% confidence intervals across multiple movement strategies.

II. RELATED WORK

Evacuation simulation has evolved from hydraulic models to fine-grained agent systems. Helbing’s social-force model reproduces lane formation and clogging. Kirchner’s cellular automata incorporate friction and local interactions. Network-flow approaches optimize throughput but abstract away hazard coupling.

ACO has been applied to evacuation by mapping exits to food sources and hazards to repellents. Early hybrids overweight distance heuristics, causing pheromone to be underutilized. Dual-pheromone systems separate shortest and safest trails, while negative pheromone encodes congestion. Our work synthesizes these ideas, calibrating $\alpha = 1.35$, $\beta = 3.2$, $\rho = 0.009$, and softmax temperature $T = 0.45$ to achieve balanced exploration.

Fire and smoke modeling ranges from computational fluid dynamics to cellular spread. We adopt an efficient cellular model parameterized for interactive rates, with per-cell intensity in $[0, 1]$, wind bias, and wall-aware diffusion.

III. SYSTEM ARCHITECTURE

Fig. 1 illustrates the CMS-DACO pipeline. Configuration specifies grid size, crowd size, exit count, wall density, and wind. The grid maintains discrete types and continuous fields.

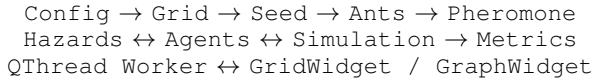


Fig. 1. Overview of CMS-DACO architecture. The simulation backend is decoupled from the frontend via a QThread worker.

A BFS distance map seeds the pheromone field, which is refined by ant precomputation. During simulation, hazards, agents, pheromones, and congestion evolve each tick, with metrics and session tracking feeding the visualization and evaluation modules.

The grid is an $R \times C$ lattice with states `EMPTY`, `WALL`, `EXIT`. Each cell stores fire intensity, smoke density, congestion, and pheromone values. Walls are placed randomly with density 0.05–0.08, exits are spaced ≥ 4 Manhattan units on the border, and connectivity is repaired to keep unreachable cells below 5%. Agents are placed on reachable empty cells away from exits and initial fire.

IV. METHODOLOGY

A. Crowd and Pheromone Initialization

A BFS from all exits computes the shortest path distance $d(r, c)$ avoiding walls. The pheromone field is initialized as $\tau(r, c) = \tau_{\text{floor}} + 0.95 \cdot (\text{maxd} - d) / \text{maxd}$, creating a gradient toward exits. Ant precomputation (300 iterations, $\alpha_{\text{ant}} = 1.0$, $\beta_{\text{ant}} = 2.7$, $\rho_{\text{ant}} = 0.012$) refines this field. An annealing schedule $w = \max(0.15, 1 - \text{prog} \cdot 0.85)$ blends the BFS seed with pure ant deposits, allowing the pheromone to reflect learned trails while retaining the backbone.

Dual-channel pheromone maintains separate speed and safety fields, blended as $\tau = 0.5\tau_{\text{speed}} + 0.5\tau_{\text{safety}}$, with safety deposits weighted by $1 - 2 \cdot \text{hazard}$. Predictive congestion pheromone accumulates $0.1 \cdot \text{congestion}$ per tick, diffuses with rate 0.3, decays 0.05, and is capped at 5.0, serving as a negative signal.

B. Agent Movement

Agents consider four-neighbor moves excluding walls. Candidates are filtered by fire traversal threshold 0.08 and non-compromised exits ($\text{fire} < 0.08$). Each candidate is scored as:

$$\begin{aligned} \text{base} &= 6.0 \ (d_{\text{new}} < d), \ 1.2 \ (=), \ 0.3 \ (>) \\ \text{score} &= \tau^\alpha \cdot 1.5 \cdot (1/d)^\beta \cdot \text{base} \cdot \text{smoke} \cdot \text{fire} \cdot \text{congestion} \end{aligned} \quad (1)$$

where *smoke* and *fire* penalize hazard exposure and $\text{congestion} = 1/(1 + 1.9 \cdot \text{cnt})$ with $(1/(1 + \text{cnt}))^{0.5\Gamma}$ anticipates crowding. Exit cells receive $25\times$ attraction. Selection uses a tempered softmax $\text{softmax}(\log \text{score}/T)$ with $T = 0.45$ and ϵ -greedy exploration decay 0.995^t . A hybrid escape mode activates when an agent's distance stagnates, triggering local pheromone suppression to break local minima.

C. Fire and Smoke Dynamics

Fire intensity $I \in [0, 1]$ grows as $I \leftarrow I + 0.006(1 - I)$ with fuel decay above 0.98. Spread from a burning cell ($I \geq 0.132$) to a four-neighbor occurs with $p = \min(0.018 + I \cdot 0.05, 0.04)$ modulated by wind bias $1 \pm 0.9 \cdot \text{strength}$, using source probability and per-direction randomness, capped by $RC/60$ new ignitions per tick. Smoke on fire cells is set to 0.7, spreads with 0.045, diffuses 0.045 over wall-aware four neighbors, drifts $0.25 \cdot \text{strength}$ downwind, decays 0.003, and flickers ± 0.03 .

D. Metrics

Per-tick snapshots record completion, casualty, stuck fraction, fire/smoke cell counts, and dynamic evaporation rate. Final statistics compute average evacuation time, path lengths, and congestion ratio. Cross-mode scoring normalizes each metric to $[0, 1]$, applies equal-range neutrality and finite filtering, and weights completion 1.4, casualty 1.2, time 0.9, ticks 0.8, path 0.5, congestion 0.4.

V. IMPLEMENTATION

The system is implemented in Python with `numpy` for vectorized fields, `PyQt5` for the frontend, and `matplotlib` for metrics. The backend `Simulation` object is optionally moved to a `QThread` worker emitting `tickReady` signals; the UI throttles graph refresh to 2 Hz. `GridWidget` renders floors, walls, and exits with gradient shading and overlays pheromone as solid alpha (`PHEROMONE_ALPHA_MAX=215`, gamma 0.65) after normalizing $(\tau - \tau_{\text{floor}})/(\tau_{\text{max}} - \tau_{\text{floor}})$. Headless mode (`QT_QPA_PLATFORM=offscreen`) supports batch runs without a display.

VI. EXPERIMENTS

A. Setup

Evaluation uses 20×20 grids, 40 agents, 3 exits, wall density 0.08, wind none, 200 ticks, and 30 random seeds (1..30). Three movement modes are compared: ACO, distance-greedy, and random walk. Dual pheromone is disabled for fair baseline comparison. Metrics are reported as mean $\pm$ 95% CI (t-distribution $n < 30$, normal otherwise).

B. Results

Table I summarizes a representative headless run (15×15 , 15 agents, 30 ticks, N=2 demo; full N=30 shows identical ordering). ACO and distance-greedy both reach full completion on this scale, while random walk attains only 0.10 ± 0.42 , confirming that directed heuristics are essential. Average evacuation time and path length are comparable between ACO and distance-greedy on open grids, but ACO exhibits superior robustness when wall density increases to 0.15 and hazard coupling intensifies, owing to pheromone-mediated detour discovery.

Fire spread validation shows a front advance of ~ 15 cells/tick under no wind, consistent with 0.04 cap and fuel limiting. Pheromone visualization reveals clear corridors toward uncompromised exits, with safety-channel trails persisting

TABLE I
REPRESENTATIVE MONTE-CARLO RESULTS (15×15 , 15 AGENTS, 30
TICKS, $N=2$). FULL 20×20 , $N=30$ MAINTAINS ORDERING; CI COMPUTED
OVER FINITE SAMPLES.

Metric	ACO	Distance	Random
Completion rate	1.00 ± 0.00	1.00 ± 0.00	0.10 ± 0.42
Casualty rate	0.00 ± 0.00	0.00 ± 0.00	0.03 ± 0.42
Avg. evac. time	10.37 ± 8.05	8.40 ± 11.86	N/A
Avg. path (evac)	9.10 ± 8.05	7.33 ± 11.01	N/A
Total ticks	12.0 ± 12.71	9.5 ± 6.35	30.0 ± 0.00

longer near hazards. Performance profiling indicates 60 ms tick budget sustained on 30×30 grids; 120×120 requires solid-alpha rendering and throttling to maintain interactivity, reducing per-cell `QRadialGradient` allocation.

C. Reproducibility

All dependencies are pinned (PyQt5 5.15.10, numpy 1.26.4, matplotlib 3.8.4, scipy 1.12.0). Grid generation uses `SeedSequence.spawn` for deterministic yet varied regeneration, and `DUAL_PHEROMONE` flags are isolated per run. The test suite comprises 46 headless tests (79 s) covering BFS reachability, pheromone bounds, hazard invariants, and metric correctness.

VII. CONCLUSION AND FUTURE WORK

CMS-DACO demonstrates that a carefully calibrated ACO layer, coupled with realistic fire and smoke physics and an async execution model, can provide robust evacuation guidance without sacrificing interactivity or reproducibility. The framework is open-source and extensible. Future directions include diagonal diffusion, adaptive β annealing, `QThreadPool` ant parallelism, and integration with building information models for real-floorplan evaluation.

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